

- 2 The human pancreas synthesises and secretes a number of different digestive enzymes. These enzymes have their effect in the small intestine.

Pancreatic lipase acts by breaking the bond between glycerol and fatty acids in triglycerides.

- (a) State the term used to describe any enzyme that is secreted to the outside of the cell where it has its effect.

..... [1]

- (b) One of the triglycerides found in olive oil is shown in Fig. 2.1.

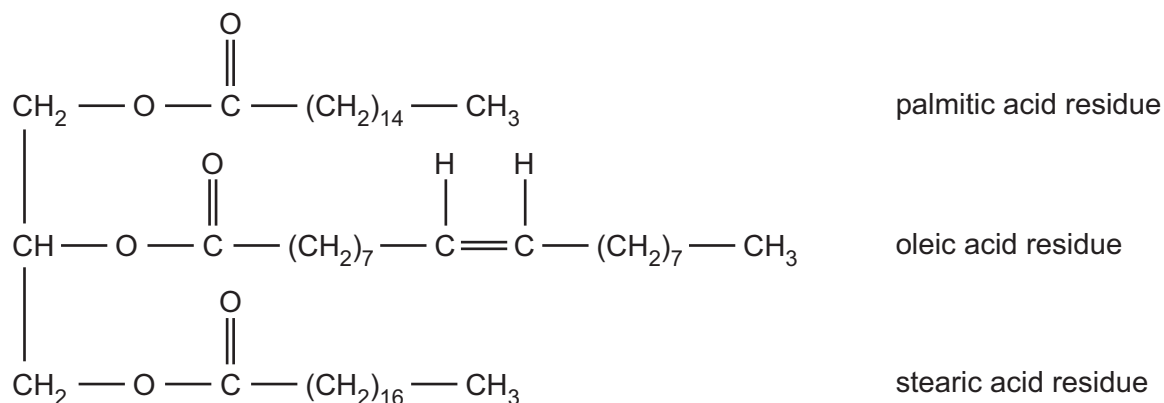


Fig. 2.1

- (i) Draw an arrow on Fig. 2.1 to show where the bond between the glycerol residue and the stearic acid residue is broken by lipase digestion. [1]
- (ii) State the type of reaction that occurs to produce stearic acid from this triglyceride **and** name the type of bond that is broken.

type of reaction

type of bond

[2]

Investigations were carried out into the activity of a lipase extracted from a strain of bacterium that lives in hot springs.

- (c) The activity of the bacterial lipase was measured at 37 °C using different concentrations of olive oil as the substrate.

The results were used to derive a Michaelis–Menten constant (K_m) of 91.76 mmol dm⁻³.

Explain what is meant by a K_m of 91.76 mmol dm⁻³.

.....

.....

.....

.....

..... [2]

- (d) Fig. 2.2 shows the results of an investigation into the effect of temperature on the activity of the bacterial lipase.

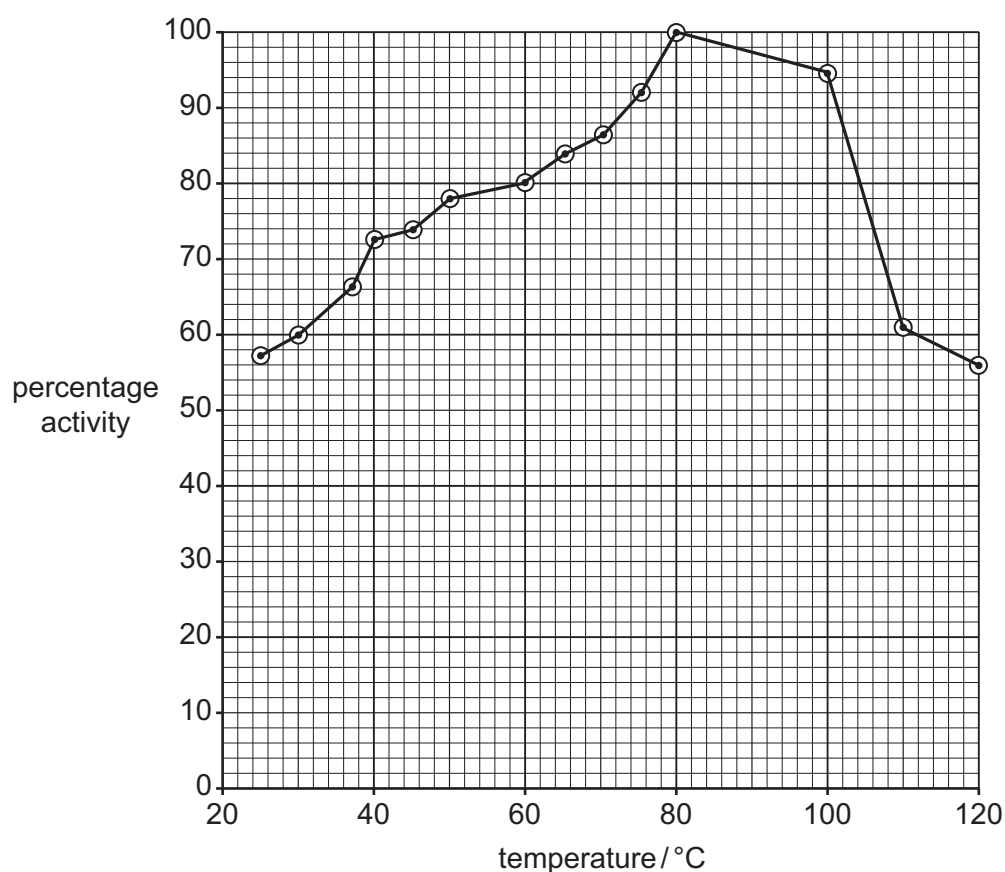


Fig. 2.2



(i) Describe **and** explain the results shown in Fig. 2.2.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) An investigation using human pancreatic lipase at the same range of temperature will produce different results from those shown in Fig. 2.2.

Predict how the trend will be different from that shown in Fig. 2.2.

.....

.....

.....

.....

..... [2]

[Total: 12]